Multiteacher Multiobjective Meta-Learning for Zero-Shot Heterogeneous Hyperspectral Band Selection

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***Abstract*—Band selection plays a crucial role in hyperspectral image (HSI) classification by removing redundant and noisy bands and retaining discriminative ones. However, most existing cross-scene band selection methods are targeted at specific homogeneous datasets. In heterogeneous datasets with different numbers of spectral bands and wavelength ranges, the previous methods exhibit limited ability to generalize to unseen target datasets. To address this issue, a novel multiteacher multiobjective meta-learning network (M^3BS) is proposed for zero-shot hyperspectral band selection and classification. In M^3BS, heterogeneous spectral information is effectively modeled through a spatial–spectral graph, enabling a generalizable graph convolutional network (GCN) to generate a dataset-agnostic base, and extract compatible meta-knowledge from multiple band selection tasks. To enhance the ability of meta-knowledge extraction, multiple band selection teachers are introduced to provide diverse, high-quality experiences. Finally, subsequent classification tasks are attached and jointly optimized with multiteacher band selection tasks through multiobjective meta-learning in an end-to-end trainable way. Multiobjective meta-learning guarantees to coordinate diverse optimization objectives automatically and adapt to various datasets simultaneously. Once the optimization is accomplished, the acquired meta-knowledge can be directly transferred to unseen datasets without any retraining or fine-tuning. Experimental results demonstrate that our proposed method outperforms state-of-the-art baselines for zero-shot hyperspectral band selection. The code is available at https://github.com/jiefeng0109/M3BS.**

***Index Terms*—Band selection, hyperspectral image (HSI), meta-learning, multiobjective learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture hundreds of narrow spectral bands, enabling a more precise analysis of land cover and finer spectral resolution than multispectral systems through continuous spectral signatures. This rich semantic information provides critical applications in environmental monitoring, agriculture, and geology [1], [2], [3], [4], [5]. It simultaneously introduces the challenges of high dimensionality, where the abundance of noisy and redundant bands can paradoxically degrade classification accuracy, alongside raising transmission bandwidth requirements and requiring exponentially large training datasets for robust generalization [6]. These constraints make dimensionality reduction essential to eliminate irrelevant spectral information while preserving discriminative features.

In HSIs, dimensionality reduction primarily involves feature extraction and band selection [7]. Unlike feature extraction, which alters original data representations, band selection retains original physical information by directly removing redundant or noisy bands [8]. For the subsequent image classification tasks, band selection optimizes band subsets to maximize classification performance.

From a transfer learning perspective, band selection methods can be categorized into four main paradigms based on the relationship between training and test data distributions, as shown in Fig. 1.

In Fig. 1(a), the methods usually rely on the independent and identically distributed assumption, which requires complete retraining for each new dataset [9]. But labeling samples is a time-consuming and labor-intensive task. There is a domain shift between different scenes, which limits the practical application of the above methods [10]. To address the limitations, an effective solution is to design a cross-scene method [11], [12], [13]. Cross-scene learning, as a transfer learning technique in remote sensing, aims to transfer the inherent correlation knowledge acquired from the source domain (SD, i.e., source scene) to the target domain (TD, i.e., target scene). As shown in Fig. 1(b), the learned representations from SD serve as domain-invariant prior knowledge, improve the band selection or classification performance in TD through parameter transfer and feature space alignment [14], [15], [16], [17], [18], i.e., domain adaption and domain generalization problem.

However, these homogeneous methods apply only when source and target scenes share the same sensor. In real-world applications, the scenes are captured by different sensors, and HSIs vary in terms of spectral range and the number of bands. To address these issues, heterogeneous methods are employed, as illustrated in Fig. 1(c) and (d). Fig. 1(c) depicts a few-shot scenario, a paradigm designed for indirect generalization. It

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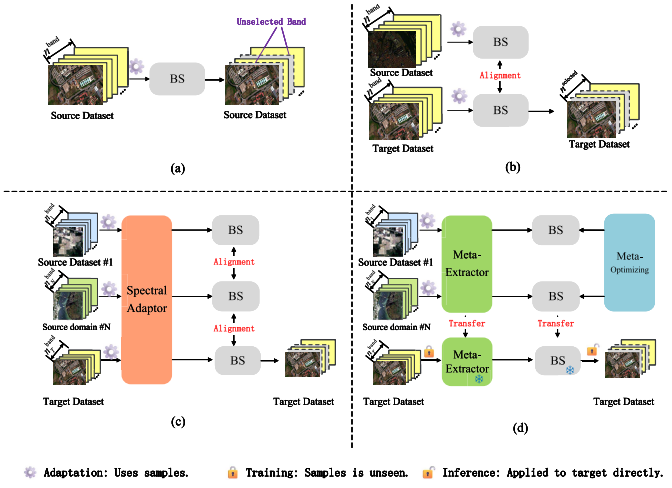


Fig. 1. Comparison of four major paradigms for hyperspectral band selection (BS). Gray squares denote unselected bands. (a) Single-scene selection, which requires retraining for each new dataset. (b) Homogeneous cross-scene selection, which adapts to sensor-extrinsic shifts between different scenes. (c) Few-shot heterogeneous selection, adapting to sensor-intrinsic shifts by using a small number of target samples. (d) Proposed zero-shot paradigm, which transfers meta-knowledge to unseen datasets without requiring any target samples.

works by rapidly adapting to a TD with a limited number of samples, often utilizing a spectral adaptor component to align the different data distributions [19], [20], [21]. In contrast, the zero-shot scenario, shown in Fig. 1(d), enables direct cross-domain transfer without requiring any samples from the TD. This is achieved by training a model to generalize across SDs, so it can be directly applied for inference on the target.

Developing the zero-shot band selection is nontrivial. Robust band selection in heterogeneous scenarios necessitates effective meta-knowledge modeling from spectral information [22]. The existing zero-shot band selection methods cannot explicitly separate domain-invariant spectral correlations from domain-specific sensor features. This entanglement leads to biased meta-knowledge that overfits SD spectral distributions, failing to generalize to unseen target sensors with divergent capture conditions [23], [24]. Furthermore, the traditional meta-learning frameworks exhibit gradient conflict when coordinating band selection and classification objectives. The opposing optimization paths of band sparsity regularization and classification loss create unstable parameter updates [25]. Unlike grid-based networks, graph convolutional networks (GCNs) excel at modeling non-Euclidean data by treating units as nodes in a topological graph [26], [27], [28]. In band selection, this allows for explicitly capturing complex spectral dependencies by representing bands as nodes and their correlations as edges, rather than simple sequences. Inspired by the above viewpoints, we leverage a generalizable GCN to explicitly decouple domain-invariant knowledge and integrate band selection and classification tasks through a unified meta-learning framework, which enhances generalization to unseen TDs.

In this article, we propose a novel multiteacher multiobjective meta-learning network (M^3BS) for zero-shot hyperspectral band selection and classification. Specifically, a spatial–spectral graph is first built to encode spectral bands of heterogeneous data into a discrete non-Euclidean space, where the vertices are regarded as bands and the edges are determined according to the spatial and spectral relationship among these bands. On this basis, a generalizable GCN is constructed as a dataset-invariant meta-knowledge extractor that decomposes the parameters to learnable dataset-agnostic bases and dataset-specific coefficients. After that, multiple classifiers are attached, including supervised and zero-shot classifiers. To guarantee a better generalization ability, several pretrained band selection teachers with a diversity ensemble strategy are designed to provide extra supervision. Finally, an uncertainty-based multiobjective meta-learning procedure is established to jointly optimize band selection and classification in an end-to-end trainable way. Once the optimization is finished, the dataset-specific coefficients can characterize unseen samples, and the acquired dataset-agnostic bases can be immediately transferred to new datasets without any retraining or fine-tuning.

The main contributions of this article are listed below.

1) The generalizable GCN can achieve meta-knowledge extraction that is compatible with various heterogeneous datasets. As the training progresses, inconsistent representations among different datasets will be synchronized by the unified meta-knowledge.
2) We incorporate multiple representative and reliable teacher models, whose multifarious experiences can maintain the generalization and adaptability of M^3BS to unseen target scenes.
3) Using the multiobjective meta-learning end-to-end optimization procedure, diverse optimization objectives can be automatically coordinated without any manual intervention, while multiple band selection tasks from multiple datasets can be simultaneously co-improved. As a result, the acquired meta-knowledge can be directly transferred from seen samples to unseen samples without any retraining or fine-tuning.
4) The proposed method enables zero-shot learning (ZSL) in hyperspectral band selection and classification. Experimental results demonstrate that it achieves the state-of-the-art effectiveness.

II. Related Work

A. Single-Scene Band Selection

Traditional single-scene hyperspectral band selection methods are categorized by label availability into supervised [29], [30], semi-supervised, and unsupervised approaches [9]. They require training separately for each new dataset and selecting band subsets. Recent advances leverage deep neural networks [31], [32], [33], [34], [35], [36], such as ternary weight (TWCNN) [37] and bandwise-independent binary architectures (BHCNN) [38] with specialized loss functions to enhance spatial modeling and nonlinear feature extraction. Moreover, unsupervised method BS-Nets [39] with L1-constrained spectral reconstruction, semi-supervised 3-D-CNN for adaptive spectral–spatial learning [40], and

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reinforcement learning frameworks formalizing selection as sequential decision processes with semi-supervised evaluators [41], collectively reduce annotation costs. Due to the specificity of different spectral ranges and land cover characteristics, these methods lack cross-dataset generalizability, which limits practical scalability and need to retrain for new datasets.

B. Cross-Scene Methods for Homogeneous HSIs Band Selection

Generally, homogeneous cross-scene HSI data are obtained by sensors of the same type, with the same spectral bands and wavelength ranges. The spectral shift is caused by factors such as location and weather between different scenes [42], [43]. To improve the generalization performance in TD, the cross-scene methods, especially domain adaptation (DA), are applied to band selection for homogeneous data with the same label and feature space of SD and TD, but different data distributions. Ye et al. [14] proposed an information gain-based method which rely on threshold segmentation and statistical calculation. Then, the RelieF [15] and Iterative RelieF [44] algorithms are extended into cross-domain versions, respectively, combined with the information of interclass neighbors. In [45], a whale optimization-based method adopts a hybrid optimization strategy to dynamically adapt to data distribution between different domains. Then, Sun et al. [46] design the cross-scene self-representative network (CSSRNet) to derive domain-invariant self-representative coefficients by employing maximum mean discrepancy (MMD). The existing methods effectively address the challenges of homogeneous data. However, they fail to transfer cross-scene information for heterogeneous hyperspectral data, which is acquired from different imaging sensors with varying numbers of spectral bands and distinct wavelength ranges.

C. Cross-Scene Methods for Few-Shot Heterogeneous HSIs

In recent years, cross-scene methods have focused on their performance on heterogeneous data. Many methods have been developed for heterogeneous HSIs classification, specifically, few-shot learning. To address this challenge, Li et al. [20] develop a deep cross-domain few-shot learning (DCFSL) framework to incorporate a conditional adversarial strategy to align domain-specific features. Moreover, Wang et al.'s framework [47] that utilizes natural images (mini-ImageNet) as a SD, acquiring meta-knowledge by VGG-based spatial feature extractors. Xiao et al. [48] aggregate abundant heterogeneous and homogeneous source data, including natural and HSIs. When it comes to band selection, the existing methods generally use multiobjective optimization algorithms to solve cross-scenario problems. They formulate band selection for each dataset as an independent optimization task. Shi et al. [21] model each task as a separate population and dynamically adjust the probability of crossover and mutation as cross-scene communication. Based on information entropy distribution, Zhang et al. [49] develop a linear mapping strategy to filter the irrelevant spectral knowledge, employ multiobjective particle swarm optimization to jointly select band subsets. However, the existing band selection methods rely on high-quality heterogeneous samples in few-shot scenarios and still face the challenges of optimizing stability and fragile generalization performance.

D. Cross-Scene Methods for Zero-Shot Heterogeneous HSIs

ZSL in hyperspectral analysis has been primarily explored in the context of image classification. This paradigm establishes a knowledge transfer framework that enables label inference for unseen categories by leveraging attribute-semantic relationships, systematically bridging prior knowledge with future recognition tasks [50]. In this domain, Pan et al. [51] proposed generating semantic label embeddings via pretrained word2vec vectors to build bidirectional correspondence with visual features. From a meta-learning perspective, Demertzis and Iliadis [52] combined multistep loss optimization with derivative order annealing, utilizing an enhanced Model-Agnostic Meta-Learning algorithm (MAML++) to extract cross-scene meta-knowledge. More recently, inspired by the generalization capabilities of Vision-Language Models, Pang et al. [53] introduced the CLIP model to HSI classification to eliminate the reliance on manual annotation.

In contrast to classification, zero-shot hyperspectral band selection remains relatively under-explored, with the primary challenge being the effective modeling of spectral heterogeneity across multisource data. Feng et al. [22] proposed MR-Selection, a two-stage meta-learning framework that aligns the directional updates of shared meta-agents across tasks. However, this initial attempt, which utilized reinforcement learning to pretrain evaluation networks via random band sampling, suffered from implicit modeling of spectral correlations and inherent training instability.

A key challenge in advancing zero-shot band selection is to learn transferable meta-knowledge that can adapt to unseen sensors with varying structural specifications and without target supervision. Meta-learning techniques those employing parameter decomposition or generation offer a promising direction. By learning to dynamically generate or modulate network weights based on task properties, these methods enable models to generalize across tasks with varying structures and input dimensions [54], [55], [56]. This characteristic offers a theoretical solution to the dimensionality mismatch problem inherent in heterogeneous hyperspectral sensors. Furthermore, ensemble learning and multiteacher distillation are widely recognized for enhancing model robustness and generalization by aggregating diverse expert knowledge [57], [58]. In the context of ZSL, where target labels are unavailable, such consensus-based mechanisms are particularly valuable for providing stable supervision signals and mitigating the inductive biases associated with individual selection criteria.

Targeting these issues, our method explicitly addresses the structural and supervisory gaps. By decoupling the GCN parameters into dataset-agnostic bases and dataset-specific coefficients, we explicitly model transferable spectral relationship knowledge. Simultaneously, by integrating multiple band selection teachers, we generate diverse, high-quality pseudo-labels to stabilize the meta-training process, thereby

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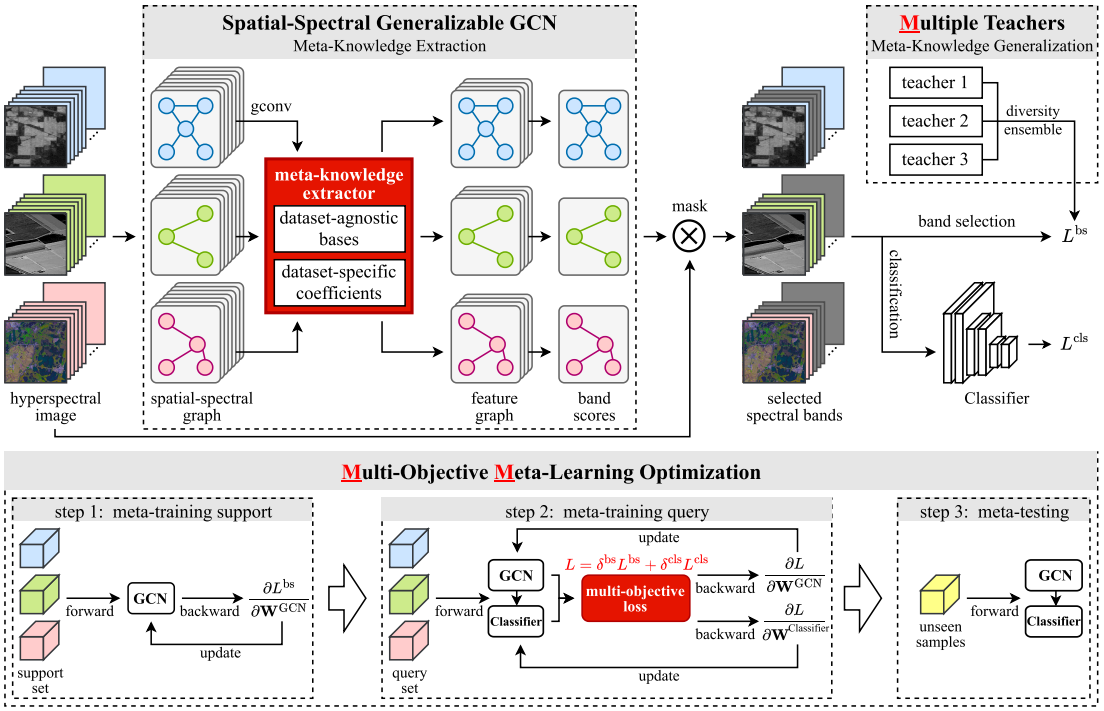


Fig. 2. Overall architecture of the proposed M^{3}BS for zero-shot hyperspectral band selection.

facilitating effective and direct generalization to unseen heterogeneous datasets.

III. Methodology

The overall architecture of the proposed M^{3}BS for zero-shot hyperspectral band selection is illustrated in Fig. 2. Specifically, the generalizable GCN with spatial–spectral graphs is constructed for dataset-agnostic band selection in Section III-A, followed by subsequent classification in Section III-B. In Section III-C, auxiliary supervision from multiple band selection teachers is introduced with a diversity ensemble strategy. Section III-E focuses on zero-shot optimization, where all the modules are integrated into a unified uncertainty-based multiobjective meta-learning framework to jointly optimize band selection and classification tasks.

A. Spatial–Spectral Generalizable GCN for Band Selection

Aiming at zero-shot band selection, we construct a generalizable GCN with spatial–spectral graphs to calculate an importance score for each band. It serves as a dataset-invariant meta-knowledge extractor to decompose the parameters into learnable dataset-agnostic bases and dataset-specific coefficients. Thus, it can synchronize inconsistent representations among different datasets, and finally learn "how to perform band selection" and achieve zero-shot inference on any new dataset.

1) Spatial–Spectral Graph: In contrast to the traditional grid structure, the graph structure can capture complex and irregular associations among individuals, which makes it particularly suitable for modeling correlation among spectral bands in HSIs. In this way, the bands and their spatial and spectral relationships are regarded as vertices and edges, respectively.

Suppose there are n^{ds} different HSI datasets $\{\mathcal{D}_1, \ldots, \mathcal{D}_{n^{\mathrm{ds}}}\}$, each of which consists of a different number of bands n^{band} but shares a common patch size $h \times w$. As for the kth dataset $\mathcal{D}_k$, the proposed spatial–spectral band graph is represented as $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{A}\}$. $\mathcal{V}$ is the set of vertices corresponding to spectral bands of HSIs, and $v_i \in \mathcal{V}$ denotes the ith spectral band. $\mathcal{E}$ is the set of edges indicating the relationships among all the spectral bands, and $e_{i,j} = (v_i, v_j) \in \mathcal{E}$ denotes the correlation between the ith and jth bands. By flattening all the pixels in a single patch, the resulting feature vector $\mathbf{x}_i \in \mathbb{R}^{hw}$ corresponds to the ith band v_i, and the feature matrix $\mathbf{X} \in \mathbb{R}^{n^{\mathrm{band}} \times hw}$ is formed by stacking together all these feature vectors. As illustrated in Fig. 3, the "blue" row of the feature matrix corresponds to the "blue" spectral band, the "green" row corresponds to the "green" spectral band, and so on.

The adjacency matrix, denoted as $\mathbf{A} \in \mathbb{R}^{n^{\mathrm{band}} \times n^{\mathrm{band}}}$, is utilized for describing the similarity between two vertices. We exploit both spatial and spectral information with a threshold for truncation θ^{adj} to calculate the adjacency matrix as follows:

$$\mathbf{A}_{i,j} = \begin{cases} \mathbf{A}^{\mathrm{spa}}_{i,j} + \mathbf{A}^{\mathrm{spec}}_{i,j}, & \text{if } \mathbf{A}^{\mathrm{spa}}_{i,j} + \mathbf{A}^{\mathrm{spec}}_{i,j} \geq \theta^{\mathrm{adj}} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

$$\mathbf{A}^{\mathrm{spa}}_{i,j} = \begin{cases} \exp\left(-\frac{1}{n^{\mathrm{band}}}|i-j|\right), & \text{if } i \neq j \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

$$\mathbf{A}^{\mathrm{spec}}_{i,j} = \begin{cases} \exp\left(-\frac{1}{hw}\left\|\mathbf{x}_i - \mathbf{x}_j\right\|_2\right), & \text{if } i \neq j \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $\mathbf{A}^{\mathrm{spa}}$ and $\mathbf{A}^{\mathrm{spec}}$ are critical to characterizing the similarity among spectral bands from different aspects. Concretely, $\mathbf{A}^{\mathrm{spa}}_{i,j}$ indicates the 1-D spatial distance between the ith and jth

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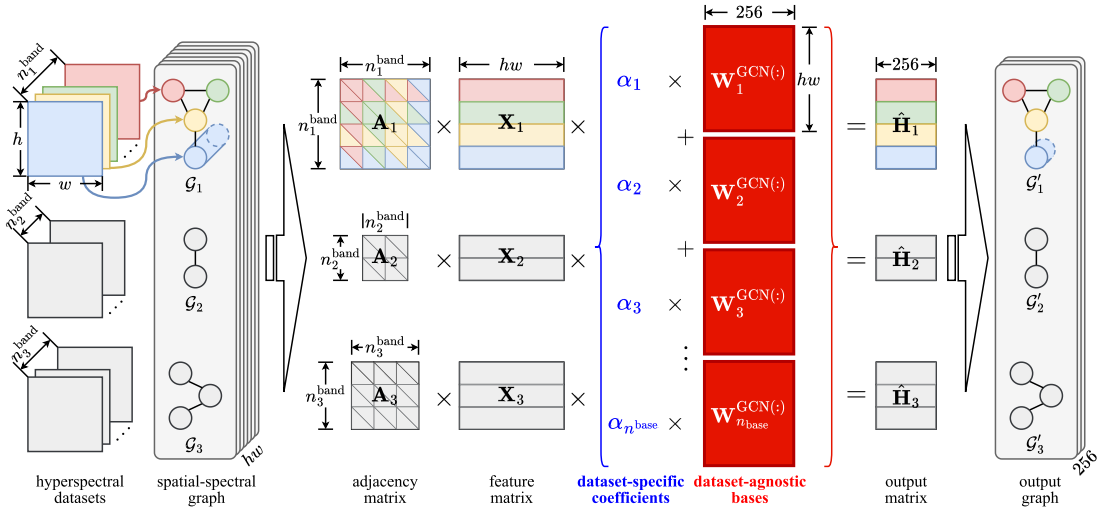


Fig. 3. First graph convolutional layer inside the generalizable GCN, comprising consecutive matrix multiplications. The weight matrix is parameterized as dataset-agnostic bases and dataset-specific coefficients. These bases are shared by different datasets, while these coefficients, which are specific to each dataset, are computed by a simple MLP network.

bands. The closer the two bands are, the closer the value is to 1. The spatial distance reflects the fact that the intensity of adjacent bands is tend to be close due to their similar light reflectance. It is worth noting that the term "spatial" in this graph construction refers to the sequential adjacency of band indices in the spectral domain, which is distinct from the spatial dimensions (height and width) of the HSI. $\mathbf{A}^{\text{spec}}_{i,j}$ indicates the l^2-norm of the difference of the feature vector $\mathbf{x}_i$ and $\mathbf{x}_j$, in the sense of the hw-dimensional Euclidean distance. As shown in (2) and (3), the Gaussian kernel function is utilized to constrain the numeric range of the two similarity matrices $\mathbf{A}^{\text{spa}}$ and $\mathbf{A}^{\text{spec}}$ between 0 and 1. Besides, we apply renormalization to the adjacency matrix $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, thereby enhancing stability in the training process.

2) Dataset-Agnostic Generalizable GCN: Meta-knowledge can be understood as generic knowledge acquired from continuous adaptation to different tasks. For meta-knowledge extraction, we propose a more effective representation of learnable parameters, which decomposes these parameters into dataset-agnostic bases and dataset-specific coefficients. These dataset-agnostic bases act as unified meta-knowledge to synchronize inconsistent representations among different datasets, while dataset-specific coefficients can effectively deal with unseen samples due to their dynamically generated characteristics.

The primary challenge in ZSL is to disentangle generalizable, transferable knowledge from dataset-specific characteristics. To achieve this, we introduce an effective parameterization for the learnable weights of the GCN, decomposing them into a set of dataset-agnostic bases and dataset-specific coefficients. This design explicitly separates the meta-knowledge from task-specific adaptation.

a) Dataset-Agnostic Bases ($\mathbf{W}_1^{GCN(:)}, \ldots, \mathbf{W}_{n^{Base}}^{GCN(:)}$): Represent the core of the meta-knowledge. These bases are *shared across all datasets* during the meta-training phase. As the model learns from diverse hyperspectral datasets, these shared bases are compeled to capture the fundamental and universal principles of spectral relationships—in essence, they learn the transferable skill of "how to perform band selection" regardless of the specific sensor or scene.

b) Dataset-Specific Coefficients ($\alpha_1, \ldots, \alpha_{n^{Base}}$): In contrast, serve as a lightweight and dynamic adaptation mechanism. They are *generated on-the-fly for each input dataset* based on its global spectral statistics, as shown in (7). These coefficients determine the appropriate linear combination of the shared bases for the current, specific dataset. This allows the model to tailor its universal knowledge to the unique properties (e.g., spectral range, number of bands) of the data it is processing.

By integrating these two components, our generalizable GCN can effectively apply its core meta-knowledge to a new, unseen dataset by simply generating a new set of coefficients for it. This process synchronizes the representations of heterogeneous datasets into a unified meta-learning framework and enables robust zero-shot inference without requiring any retraining.

Based on the abstraction of the spatial–spectral band graph, a spatial–spectral generalizable GCN is constructed for band selection in M^{3}BS, comprising two graph convolutional layers with batch normalization (BN) as shown in the following equations:

$$\hat{\mathbf{H}} = \text{ReLU}\left(\text{BN}\left(\widetilde{\mathbf{D}}^{-\frac{1}{2}}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{X}\mathbf{W}^{\text{GCN}(1)}\right)\right) \tag{4}$$

$$\hat{\mathbf{s}} = \sigma\left(\text{BN}\left(\widetilde{\mathbf{D}}^{-\frac{1}{2}}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-\frac{1}{2}}\hat{\mathbf{H}}\mathbf{W}^{\text{GCN}(2)}\right)\right) \tag{5}$$

where $\widetilde{\mathbf{A}}$ and $\widetilde{\mathbf{D}}$ are the renormalized forms of the adjacency matrix and the degree matrix, respectively, $\hat{\mathbf{H}} \in \mathbb{R}^{n^{\text{band}} \times 256}$ denotes the output feature matrix of the first graph convolutional layer, $\hat{\mathbf{s}} \in [0,1]^{n^{\text{band}}}$ indicates the importance score of each band for determining which subset should be selected, $\mathbf{W}^{\text{GCN}(1)} \in \mathbb{R}^{hw \times 256}$, $\mathbf{W}^{\text{GCN}(2)} \in \mathbb{R}^{256}$ are learnable parameters, which will be updated by gradient descent during training. The output dimensionality of each vertex in the first layer is 256.

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The second layer is normalized by a Sigmoid function $\sigma(\cdot)$ to ensure that the score of each band falls within $[0, 1]$.

Instead of static forward propagation, we parameterize the above-mentioned graph convolutional kernels $\mathbf{W}^{\mathrm{GCN}(1)}$ and $\mathbf{W}^{\mathrm{GCN}(2)}$ as linear combinations of n^{base} bases

$$\mathbf{W}^{\mathrm{GCN}(:)} = \alpha_1 \mathbf{W}_1^{\mathrm{GCN}(:)} + \cdots + \alpha_{n^{\mathrm{base}}} \mathbf{W}_{n^{\mathrm{base}}}^{\mathrm{GCN}(:)} \tag{6}$$

where $\mathbf{W}_1^{\mathrm{GCN}(:)}, \ldots, \mathbf{W}_{n^{\mathrm{base}}}^{\mathrm{GCN}(:)}$ are learnable basis matrices. During training, these bases naturally tend to become relatively orthogonal to each other. This occurs implicitly without an enforced constraint, as the model minimizes redundancy by optimizing each basis to capture distinct and complementary aspects of spectral relationships. $\alpha_1, \ldots, \alpha_{n^{\mathrm{base}}}$ are linear combination coefficients from an average pooling layer, a fully connected layer, and a sigmoid activation function

$$\alpha_1, \ldots, \alpha_{n^{\mathrm{base}}} = \sigma\left(\frac{1}{n^{\mathrm{band}}} \mathbf{1}^\top \mathbf{X} \mathbf{W}^{\mathrm{FC}}\right) \tag{7}$$

where $1/(n^{\mathrm{band}})\mathbf{1}^\top \in \mathbb{R}^{1 \times n^{\mathrm{band}}}$ calculates the mean along each column of the feature matrix $\mathbf{X}$, $\mathbf{W}^{\mathrm{FC}} \in \mathbb{R}^{hw \times n^{\mathrm{base}}}$ maps the pooled feature matrix into n^{base} scalars corresponding to n^{base} parameter bases. Meta-knowledge is explicitly represented as dataset-agnostic bases, while dataset-specific coefficients can be used to characterize unique attributes of different datasets.

According to the extracted score vector $\hat{\mathbf{s}}$ for all the bands, it is possible to determine which subset should be selected. Generally, the number of selected bands is fixed as a hyperparameter n^{sband} by users in advance ($n^{\mathrm{sband}} < n^{\mathrm{band}}$) [38]. To make it available to attach the following classification network, a binarization operator is devised to convert scores into a band mask:

$$\theta^{\mathrm{band}} = \left\{\hat{\mathbf{s}}_{o_1}, \hat{\mathbf{s}}_{o_2}, \ldots, \hat{\mathbf{s}}_{o_{n^{\mathrm{band}}}}\right\}_{n^{\mathrm{sband}}} \tag{8}$$

$$\hat{\mathbf{M}} = \phi\left(\hat{\mathbf{s}}, \theta^{\mathrm{band}}\right) = \begin{cases} 1, & \text{if } \hat{\mathbf{s}}_i \geq \theta^{\mathrm{band}} \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

Specifically, the scores of all the bands are first sorted in descending order $o_1, o_2, \ldots, o_{n^{\mathrm{band}}}$, and the threshold θ^{band} is taken to be the n^{sband}th score. Then, a binarization operator $\phi(\cdot)$ is applied to extract a band mask $\hat{\mathbf{M}} \in \{0, 1\}^{n^{\mathrm{band}}}$ by providing a conditional comparative constraint, where the corresponding value is set to 0 when the score is smaller than or equal to the threshold. Finally, this band mask is multiplied by the HSIs as the input of the classifier.

B. Image Classifier

To evaluate band subset combinations, we construct two types of classifiers, which are suitable for different classification requirements. The zero-shot classifier can unify band selection and classification in the ZSL paradigm, and the supervised classifier serves as the same paradigm for comparison with the other band selection methods.

1) Zero-Shot Classifier: We construct the latest image classifier SPECIAL [53] to achieve sample-free constraints in TD. Specifically, this method bridges the modality gap by generating pseudo-labels from spectrally interpolated RGB images via CLIP [59], and subsequently refines them to train the spectral classifier in a noise-robust manner. First, in the CLIP-based classification stage, spectral interpolation is performed on the original HSIs to generate the pseudo-labels and the corresponding confidence scores. For objects of varying scales, the resolution scaling (RS) strategy is adopted to improve the quality of pseudo-labels. Second, in the noise label learning stage, the CLIP-derived pseudo-labels guide the training of the classifier on the HSIs with spectral dimensionality reduction, where class-specific feature distributions are constructed by exploiting vision-language alignment priors from the pseudo-labels.

Based on the above two stages, the final pseudo-labels $\tilde{y}(x)$ are calculated as follows:

$$\tilde{y}_i(x) = \frac{\sum_{m=1}^{M} N(x|\mu_{im}, \sigma_{im})}{\sum_{i=1}^{n^{\mathrm{class}}} \sum_{m=1}^{M} N(x|\mu_{im}, \sigma_{im})} \tag{10}$$

where μ_{im} and σ_{im} represent the mean and covariance of the ith class generated by the mth Gaussian mixture model. Obviously, M represents the number of sets. Finally, the compound classification loss in the following equation is used to train the classifier:

$$\mathcal{L}^{\mathrm{cls}} = -\frac{1}{n^{\mathrm{batch}}} \sum_{i=1}^{n^{\mathrm{batch}}} \mathbf{1}\{\tilde{y}_i\} \log \hat{\mathbf{y}}_i. \tag{11}$$

2) Supervised Classifier: Suppose that there are n^{class} categories in the kth HSI dataset. We employ a CNN-based framework for the classification task, which is designed to process HSI patches. With the HSI patch $x \in \mathbb{R}^{n^{\mathrm{band}} \times h \times w}$ as the input, the probability distribution $\hat{\mathbf{y}} \in [0, 1]^{n^{\mathrm{class}}}$ can be predicted by the kth CNN with parameters $\mathbf{W}_k^{\mathrm{CNN}}$.

Following the settings of embedding-based band selection methods, the classification loss for a single batch is defined as a multiclass cross-entropy loss as shown in (12). $y_i \in \{1, 2, \ldots, n^{\mathrm{class}}\}$ denotes which category the center pixel of the ith patch actually belongs to, and $\mathbf{1}\{\cdot\}$ is a vectorized indicator function to perform one-hot encoding over scalars

$$\mathcal{L}^{\mathrm{cls}} = -\frac{1}{n^{\mathrm{batch}}} \sum_{i=1}^{n^{\mathrm{batch}}} \mathbf{1}\{y_i\} \log \hat{\mathbf{y}}_i. \tag{12}$$

C. Multiteacher Diversity Ensemble

As a typical preprocessing procedure for downstream tasks, band selection does not involve any explicit ground truths (G.T.) like other fully supervised tasks, such as image classification. Thus, it is usually formalized as an unsupervised task [39] or evaluated by the performance of an auxiliary classifier [60]. These band selection methods either produce poor performance or consume a significant amount of training time. Moreover, they fail to generalize to unseen samples, which makes it challenging to achieve zero-shot hyperspectral band selection.

In addition to the classification loss in Section III-B, we exploit diverse high-quality experiences from multiple band selection teachers, generating an auxiliary loss function. This extra supervision not only facilitates generalization and adaptability for the previous meta-knowledge but also accelerates

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the convergence procedure and alleviates the training time consumption.

Specifically, multiple representative and reliable teachers from filter-based, wrapper-based, and embedding-based methods, respectively, are pretrained to prepare for high-quality experiences in advance. Considering how to make a balance among multiple teachers, we formulate a diversity ensemble strategy to choose the spectral bands according to their popularity. After the band subsets are selected by these teachers, respectively, a counting function $\mathrm{cnt}(i)$ can be defined to denote the number of received votes of the ith band. As stated in (13) and (14), we can obtain a band sequence and take the top n^{sband} bands as the G.T. by sorting these bands by the number of received votes in descending order

$$\mathbb{S} = \left\{ \underbrace{i_1, \ldots,}_{\mathrm{cnt}(i)=3} \underbrace{i_7, i_8, \ldots\ldots,}_{\mathrm{cnt}(i)=2} \underbrace{i_{22}, i_{23}, \ldots\ldots\ldots,}_{\mathrm{cnt}(i)=1} \right\}_{1,\ldots,n^{\mathrm{sband}}} \tag{13}$$

$$\mathbf{s}_i = \begin{cases} 1, & \text{if } i \in \mathbb{S} \\ 0, & \text{otherwise} \end{cases} \tag{14}$$

where $\mathbf{s}$ denotes the label for directly supervising the band selection task. For bands that receive an equal number of votes at the selection threshold, a deterministic random subset is chosen from the tied bands to complete the set of n^{sband} selected bands. This ensures both fairness and reproducibility across experimental runs. With this ensemble strategy, multiple teachers can be integrated into a more powerful teacher. Thanks to the various optimizing directions from different teachers, the lack of generalization can be effectively alleviated, and this kind of diversity is beneficial to generalizing better to unseen samples in ZSL.

The band selection loss for the entire batch is defined as a multilabel binary cross-entropy loss of the continuous importance scores $\hat{\mathbf{s}} \in [0,1]^{n^{\mathrm{band}}}$ from the GCN and the discrete selected bands $\mathbf{s} \in \{0,1\}^{n^{\mathrm{band}}}$ from the integrated teacher

$$\mathcal{L}^{\mathrm{bs}} = \mathbf{s} \log \hat{\mathbf{s}} + (\mathbf{1} - \mathbf{s}) \log (\mathbf{1} - \hat{\mathbf{s}}). \tag{15}$$

D. Uncertainty-Based Multiobjective Learning

Both the classification loss in Section III-B and the selection loss in Section III-C are beneficial for the expected band selection task. It inspires us to simultaneously optimize them with an integrated loss function. Intuitively, combining multiple losses could be simply accomplished by applying a weighted sum and constituting the overall loss function $\mathcal{L}_{\mathrm{vanilla}}$ as follows:

$$\mathcal{L}_{\mathrm{vanilla}} = \lambda^{\mathrm{bs}} \mathcal{L}^{\mathrm{bs}} + \lambda^{\mathrm{cls}} \mathcal{L}^{\mathrm{cls}} \tag{16}$$

where $\lambda^{\mathrm{bs}} \in [0,1]$ and $\lambda^{\mathrm{cls}} \in [0,1]$ are relative weights for the band selection loss and the classification loss, respectively. In n^{ds} different HSI datasets, there are $2n^{\mathrm{ds}}$ extra hyperparameters needed to be tuned and denoted as $\{\lambda_1^{\mathrm{cls}}, \ldots, \lambda_{n^{\mathrm{ds}}}^{\mathrm{cls}}, \lambda_1^{\mathrm{bs}}, \ldots, \lambda_{n^{\mathrm{ds}}}^{\mathrm{bs}}\}$. Actually, the performance of the band selection task is heavily sensitive to the proportion of these weights [61]. However, it is difficult to tune these hyperparameters manually.

It is more ideal to allow the loss weights to be updated together with network parameters during the training process. Inspired by [62], a multiobjective learning approach is proposed, which weighs multiple loss functions by considering the uncertainty of each task. Kendall et al. [62] stated that the accidental error between model outputs and sample labels can be modeled as a homoscedastic uncertainty, which allows us to decompose the multiobjective loss into the product of multiple likelihoods. By estimating these two tasks with a Sigmoid likelihood $\sigma(\cdot)$ and a Softmax likelihood $\mathrm{Softmax}(\cdot)$, respectively, an uncertainty-based minimization objective can be defined as follows:

$$\begin{aligned} \mathcal{L} &= -\log \sigma\left(\hat{\mathbf{s}}_k;\ \mathbf{W}^{\mathrm{GCN}}, \lambda^{\mathrm{bs}}\right) \\ &\quad \cdot \mathrm{Softmax}\left(\hat{\mathbf{y}}_k;\ \mathbf{W}^{\mathrm{GCN}}, \mathbf{W}_k^{\mathrm{Classifier}}, \lambda^{\mathrm{cls}}\right) \\ &\propto \lambda^{\mathrm{bs}} \mathcal{L}^{\mathrm{bs}} + \lambda^{\mathrm{cls}} \mathcal{L}^{\mathrm{cls}} + \log \sqrt{\frac{1}{\lambda^{\mathrm{bs}}}} + \log \sqrt{\frac{1}{\lambda^{\mathrm{cls}}}} \end{aligned} \tag{17}$$

where λ^{bs} and λ^{cls} are two weights corresponding to the above-mentioned two objectives. As λ^{bs} or λ^{cls} decreases, $1/(\lambda^{\mathrm{bs}})^2$ or $1/(\lambda^{\mathrm{cls}})^2$ increases, which brings a greater influence to the corresponding loss function. These relative weights are also discouraged from decreasing excessively by the last regularization term $\log(1/\lambda^{\mathrm{bs}})^{1/2}$ and $\log(1/\lambda^{\mathrm{cls}})^{1/2}$. The weights λ^{bs} and λ^{cls} are both initialized to 1.0, which zeroes the regularization terms in (17) at the start of training and enables stable gradient updates until task-specific uncertainties are learned.

By applying (17) rather than (16) to optimize multiple objectives, massive hyperparameter tuning costs can be avoided since the weights are determined automatically by gradient descent without any manual intervention, and different training stages can be adapted since the weights are updated dynamically as training progresses.

E. Multiobjective Meta-Learning for Zero-Shot Optimization

When it comes to zero-shot inference, we try to figure out how to perform hyperspectral band selection on a new dataset without any of its samples during training. To achieve zero-shot optimization, we construct a multiobjective meta-learning procedure to formalize an intuitive trial-and-error process. It makes our framework not only suitable for both the band selection and classification objectives, but also applicable to various datasets with different characteristics, thereby ensuring maximum compatibility.

Following how previous works preprocess data for meta-learning [63], [64], [65], we divide HSI datasets into meta-training and meta-testing tasks. Our goal is to achieve meta-knowledge extraction from meta-training tasks and apply it to meta-testing tasks. Specifically, let HSI datasets $\mathcal{D}_1^{\mathrm{train}}, \ldots, \mathcal{D}_{n^{\mathrm{ds}}}^{\mathrm{train}}$ be different meta-training tasks for training, $\mathcal{D}^{\mathrm{test}}$ be the meta-testing task for zero-shot inference. What we want to achieve is to directly infer on $\mathcal{D}^{\mathrm{test}}$ after training on $\mathcal{D}_1^{\mathrm{train}}, \ldots, \mathcal{D}_{n^{\mathrm{ds}}}^{\mathrm{train}}$, and none of the samples in $\mathcal{D}^{\mathrm{test}}$ are provided during the training phase. Besides, each meta-training task (i.e., dataset) $\mathcal{D}_k^{\mathrm{train}}$ is further divided into a support set $\mathcal{D}_k^{\mathrm{spt}}$ and a query set $\mathcal{D}_k^{\mathrm{qry}}$ according to a predefined percentage.

The overall procedure of M^{3}BS is described elaborately in Algorithm 1. Inspired by model-agnostic meta-learning [66], the whole meta-training process can be viewed as a trial-and-error process in Fig. 4, where the support stage

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Algorithm 1 M^{3}BS

Input: $\mathcal{D}_1^{\text{train}}, \ldots, \mathcal{D}_{n^{\text{ds}}}^{\text{train}}, \mathcal{D}^{\text{test}}$: meta-training and meta-testing tasks; α: learning rate; β: meta-learning rate; T: number of epochs.
Output: $\mathbf{W}^{\text{GCN}(:)}, \mathbf{W}_1^{\text{Classifier}}, \ldots, \mathbf{W}_{n^{\text{ds}}}^{\text{Classifier}}$: well-train parameters.

Initialize relative weights: $\lambda^{\text{bs}}|_{t=1} \leftarrow 1, \lambda^{\text{cls}}|_{t=1} \leftarrow 1$
for epoch $t = 1, \ldots, T$ **do**
  for each meta-training task $\mathcal{D}_k^{\text{train}}$ in $\{\mathcal{D}_1^{\text{train}}, \ldots, \mathcal{D}_{n^{\text{ds}}}^{\text{train}}\}$ **do**
    Split samples for support and query: $\mathcal{D}_k \rightarrow \{\mathcal{D}_k^{\text{spt}}, \mathcal{D}_k^{\text{qry}}\}$
    Temporarily duplicate parameters: $\mathbf{W}_k^{\text{GCN}} \leftarrow \mathbf{W}^{\text{GCN}}$
    for each batch in $\mathcal{D}_k^{\text{spt}}$ **do**
      Flatten all pixels into a feature matrix $\mathbf{X}$
      Characterize an adjacency matrix $\mathbf{A}$ by (1)-(3)
      Build a spatial-spectral graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{A}\}$
      Assign importance scores $\hat{\mathbf{s}}$ with $\mathbf{W}^{\text{GCN}}$ by (4)-(7)
      Vote a band subset $\mathbf{s}$ from teachers by (13)-(14)
      Measure the band selection loss $\mathcal{L}^{\text{bs}}$ by (15)
      Backpropagate gradients to obtain $\partial\mathcal{L}^{\text{bs}}/\partial\mathbf{W}_k^{\text{GCN}}$
      Update $\mathbf{W}_k^{\text{GCN}}$ with gradient descent: $\mathbf{W}_k^{\text{GCN}}|_{t+1} = \mathbf{W}_k^{\text{GCN}}|_t + \alpha \cdot \partial\mathcal{L}^{\text{bs}}/\partial\mathbf{W}_k^{\text{GCN}}|_t$
    end for
    for each batch in $\mathcal{D}_k^{\text{qry}}$ **do**
      Obtain $\hat{\mathbf{s}}$ and $\mathcal{L}^{\text{bs}}$ by following the same steps as above
      Binarize $\hat{\mathbf{s}}$ into a band mask $\hat{\mathbf{M}}$ by (8)-(9)
      for each patch x_i and category y_i in this batch **do**
        Mask x_i with $\hat{\mathbf{M}}$ by multiplication: $x_i \leftarrow x_i \cdot \hat{\mathbf{M}}$
        Classify x_i into a probability vector $\hat{\mathbf{y}}_i$ with $\mathbf{W}_k^{\text{Classifier}}$
      end for
      Measure the zero-shot or supervised classification loss $\mathcal{L}^{\text{cls}}$ by (11)-(12)
      Measure the multi-objective loss $\mathcal{L}$ by (17)
      Backpropagate gradients to obtain $\partial\mathcal{L}/\partial\lambda^{\text{bs}}$, $\partial\mathcal{L}/\partial\lambda^{\text{cls}}$, $\partial\mathcal{L}/\partial\mathbf{W}_k^{\text{GCN}}$ and $\partial\mathcal{L}/\partial\mathbf{W}_k^{\text{Classifier}}$
      Accumulate $\partial\mathcal{L}/\partial\mathbf{W}_k^{\text{GCN}}$ for $\mathbf{W}_k^{\text{GCN}}$
      Update λ^{bs}, λ^{cls} and $\mathbf{W}_k^{\text{Classifier}}$ with gradient descent:
      $\lambda^{\text{bs}}|_{t+1} = \lambda^{\text{bs}}|_t + \alpha \cdot \partial\mathcal{L}/\partial\lambda^{\text{bs}}|_t$
      $\lambda^{\text{cls}}|_{t+1} = \lambda^{\text{cls}}|_t + \alpha \cdot \partial\mathcal{L}/\partial\lambda^{\text{cls}}|_t$
      $\mathbf{W}_k^{\text{Classifier}}|_{t+1} = \mathbf{W}_k^{\text{Classifier}}|_t + \alpha \cdot \partial\mathcal{L}/\partial\mathbf{W}_k^{\text{Classifier}}|_t$
    end for
    Average $\partial\mathcal{L}/\partial\mathbf{W}_k^{\text{GCN}}$ over all batches for $\mathbf{W}_k^{\text{GCN}}$
    Summarize $\partial\mathcal{L}/\partial\mathbf{W}_k^{\text{GCN}}$ to $\partial\mathcal{L}/\partial\mathbf{W}^{\text{GCN}}$ for $\mathbf{W}^{\text{GCN}}$: $\partial\mathcal{L}/\partial\mathbf{W}^{\text{GCN}} \leftarrow \partial\mathcal{L}/\partial\mathbf{W}^{\text{GCN}} + \partial\mathcal{L}/\partial\mathbf{W}_k^{\text{GCN}}$
  end for
  Update $\mathbf{W}^{\text{GCN}}$ with gradient descent for meta-learning: $\mathbf{W}^{\text{GCN}}|_{t+1} = \mathbf{W}^{\text{GCN}}|_t + \beta \cdot \partial\mathcal{L}/\partial\mathbf{W}^{\text{GCN}}$
end for

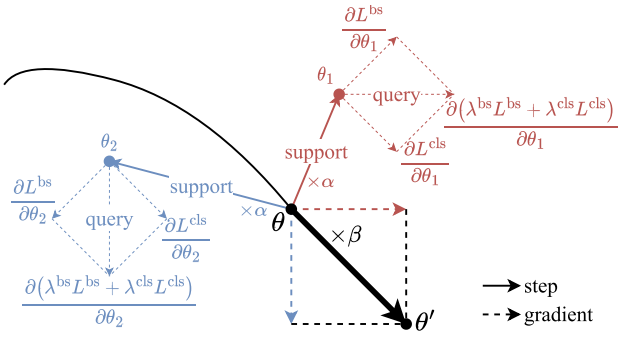


Fig. 4. Multiobjective meta-learning procedure for zero-shot optimization, where different colors correspond to different HSI datasets.

makes an attempt, and the query stage verifies the effectiveness of this attempt. Significantly different from [66], instead of applying the same loss function for both stages, we optimize the multiobjective loss in Section III-D in the query stage, thereby utilizing the stronger supervision from classification to more effectively verify whether the attempt made in the support stage is reasonable. In the support stage, only the band selection loss $\mathcal{L}^{\text{bs}}$ is involved for a tentative update step for the temporary parameters $\mathbf{W}_k^{\text{GCN}}$. In the query stage, the multiobjective loss $\mathcal{L}$ is used to verify whether the previous step is beneficial for classification or not. For verification, only the derivatives of the loss with respect to the parameters $\partial\mathcal{L}/\partial\mathbf{W}_k^{\text{GCN}}$ are calculated, and the parameters $\mathbf{W}_k^{\text{GCN}}$ are not actually updated. After iterating over all n^{ds} meta-training tasks, all these derivatives are translated to the original parameters $\mathbf{W}^{\text{GCN}}|_t$ for gradient descent, leading to a more generalizable version $\mathbf{W}^{\text{GCN}}|_{t+1}$. During meta-testing, discriminative and informative band subsets can be obtained by applying the compatible meta-knowledge extracted by the generalizable GCN to perform zero-shot inference.

IV. Experimental Results and Analysis

In this section, quantitative statistical experiments and qualitative analysis are conducted over multiple commonly used hyperspectral datasets to validate the effectiveness and efficiency of our method. Detailed descriptions of these datasets and experimental settings are provided in Sections IV-A and IV-B, respectively.

A. Dataset Description

Our crafted experiments involve a total of 6 hyperspectral datasets, namely, Indian Pines, Pavia University, University of Houston, Salinas, Washington DC Mall, and Kennedy Space Center.

1) Indian Pines: Being a publicly recognized hyperspectral band selection dataset, Indian Pines was collected in 1992 by the Airborne Visible InfraRed Imaging Spectrometer (AVIRIS) over a landscape in Indiana, USA. Technically, this dataset consists of an HSI with 145×145 pixels.

With a wavelength range of 0.4–2.5 μm, the airborne visible infrared imager captures a total of 224 spectral reflectance bands, and discards 24 bands that cover the region of water absorption. A false-color image composited by the bands 50, 27, and 17 is shown in Fig. 5(a).

2) Pavia University: As a partial scene of Pavia, northern Italy, this dataset was acquired by the German Airborne Reflective Optical Spectral Imager Hyperspectral Data Spectral Imager (ROSIS) in 2003. It contains 115 consecutive bands, 12 bands of which are previously removed due to noise interference. The HSI has a spatial resolution of 610×340 and contains a total of 2 207 400 pixels, including a large number of background pixels. Fig. 5(b) shows the extracted bands 53, 31, and 8.

3) University of Houston: This dataset was obtained by the ITRES CASI-1500 sensor over a campus and its neighboring urban area and served as a contest for the 2013 IEEE GRSS Data Fusion Competition. It is an HSI of size 349×1905 and contains 144 bands in the spectral range from 364 to 1046 nm. Fig. 5(c) displays a false-color image with pixel-level G.T. composed of the bands 28, 45, and 65.

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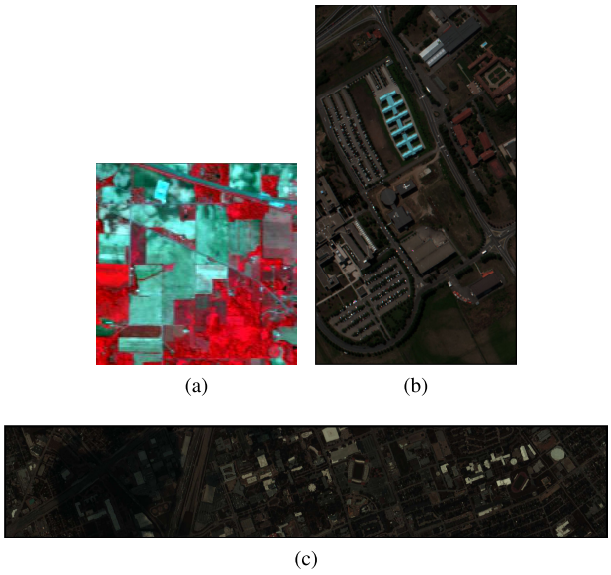


Fig. 5. False-color images composited by three different spectral bands from (a) Indian Pines, (b) Pavia University, and (c) University of Houston.

4) Salinas: This scene was gathered by the AVIRIS sensor over Salinas Valley, California, with a high spatial resolution of 512×217. Same as Indian Pines described before, only 204 spectral bands are preserved by discarding 20 bands covering water absorption. This dataset incorporates 16 categories, such as grapes, stubble, and celery.

5) Washington DC Mall: This 191-band dataset was taken by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor with a wavelength range of 0.4–2.4 μm. The HSI contains 750×307 pixels, each of which is classified into one of nine categories, such as roof and grass.

6) Kennedy Space Center: Generally abbreviated as KSC, this dataset was collected by the AVIRIS sensor in 1996. The spatial resolution of this dataset is 512×616. The number of spectral bands is also cut down from 224 to 176 by removing both water absorption and low SNR bands. To achieve pixel-level classification, land cover objects are grouped into 13 categories, e.g., scrub, slash pine, and salt marsh.

Following the same practice of previous works, all samples are randomly divided into a training set and a testing set at a ratio of 5:95 for Indian Pines and University of Houston, and a ratio of 3:97 for Pavia University. All training sets are further subdivided into support sets and query sets for meta-learning optimization at a unified ratio of 3:7, which guarantees that all categories are present during training.

B. Experimental Setups

To more convincingly demonstrate the effectiveness of our algorithm in various aspects, up to 12 existing band selection algorithms are chosen for performance comparison, including mRMR [67], BS-Nets [39], GCSR-BS [35], TGSR [68], DGLAnet [69], TSC [70], SICNN [31], ABCNN [33], CDWOASA [45], CSSRnet [46], and MR-Selection [22]. Among all these methods, BS-Nets, GCSR-BS, TGSR, DGLAnet, and TSC are unsupervised single-scene methods. The mRMR, SICNN, and ABCNN are supervised single-scene methods. CDWOASA and CSSRnet are cross-scene methods for homogeneous HSIs, while MR-Selection is for heterogeneous HSIs.

The methods, mRMR, GCSR-BS, TGSR, DGLAnet, and TSC, require additional classifiers, which follow their default parameter settings. Here, a SVM classifier is attached by grid searching c and γ in the range of {1, 10, 100, 1000, 10 000} and {0.01, 0.1, 1, 10, 100}, respectively. For BS-Nets with a trainable CNN, the learning rate is fixed to 0.001, the number of training epochs is set to 500, and the regularization coefficient is set to 0.02. For deep learning-based methods, BS-Nets, GCSR-BS, SICNN, and ABCNN, the hyperparameters such as batch size, learning rate, and training epochs are determined either with a trial-and-error procedure or by following the settings of the original paper.

For our M^{3}BS, both the learning rate α for optimizing classification and the meta-learning rate β for optimizing band selection are set to 0.001, and further adopt an exponential scheduler to decay themselves every epoch. Both the generalizable GCN and the CNN-based image classifiers are optimized with a gradient-based stochastic algorithm, Adam [71]. The batch size n^{batch} is set to 128, and the number of epochs T is set to 400. For any batch of any dataset, the spatial size of the HSI patch is fixed to 33×33. By controlling the truncation threshold θ^{adj} for the adjacency matrix $\mathbf{A}$, the number of edges is limited to less than 1000 to prevent over-densification. The number of dataset-agnostic bases for parameterizing the graph convolutional kernels is set to 3. After extracting importance scores for each band, the number of selected bands n^{sband} is fixed to 20 for a fair comparison among different datasets.

From a diversity perspective, DDCNN, SICNN, and TWCNN, which represent filter-based, wrapper-based, and embedding-based methods, respectively, are chosen to be our teachers. To strictly adhere to the zero-shot protocol, the pretraining of these teacher models is conducted exclusively on the meta-training datasets (i.e., Salinas, Washington DC Mall, and Kennedy Space Center), ensuring complete isolation from the meta-testing TDs. During this phase, we strictly follow the original experimental settings and loss functions proposed in their respective papers to optimize each teacher model. Consequently, each teacher independently identifies discriminative band subsets based on the SD data. These subsets are subsequently aggregated via the diversity ensemble strategy to generate the binary G.T. $\mathbf{s}$. This isolation strategy effectively prevents data leakage, ensuring that the supervisory signals for the meta-learning framework are derived solely from the SDs.

For the sake of reproducibility, detailed experimental procedures and well-commented implementations in PyTorch [72]. All the experiments are conducted on a computer with an AMD Ryzen 5950X CPU and an Nvidia GeForce RTX 3090 GPU.

C. Single-Scene Performance

In this section, to preliminarily confirm the effectiveness of our proposed network structure, we first verify its classification performance in the standard single-scene setting. This scenario is distinct from ZSL, as training and testing occur on the

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TABLE I
PERFORMANCE OF SUPERVISED CLASSIFICATION IN A SINGLE-SCENE SETTING

Method	Indian Pines			Pavia			Houston		
	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa
mRMR [67]	71.7±0.6	63.1±3.4	67.6±0.7	80.5±0.5	73.8±0.3	73.5±0.6	82.2±0.2	82.0±0.2	80.8±0.3
BS-Nets [39]	68.7±0.9	59.9±1.8	64.3±1.0	79.8±0.9	71.6±2.0	72.2±1.4	83.3±0.5	82.3±0.7	82.0±0.6
GCSR-BS [35]	68.3±2.5	74.6±1.1	71.0±1.2	82.6±0.5	86.6±0.4	82.1±0.6	88.1±0.5	87.1±0.5	87.1±0.5
TGSR [68]	74.3±0.6	73.8±0.6	70.7±0.7	86.7±0.5	86.1±0.5	82.0±0.7	88.5±0.6	88.2±0.7	87.6±0.6
DGLAnet [69]	81.7±0.4	79.9±1.0	79.1±0.1	91.9±0.2	89.3±0.5	89.3±0.1	91.4±0.3	90.3±0.6	90.8±0.4
TSC [70]	71.4±0.7	67.3±3.2	67.3±0.7	83.9±0.3	80.8±0.7	78.5±0.4	83.8±0.3	83.6±0.2	82.9±0.3
SICNN [31]	87.6±1.0	82.7±2.6	85.9±1.1	95.3±0.9	93.1±0.7	93.7±0.7	93.8±0.7	92.7±1.7	93.4±0.9
ABCNN [33]	81.5±1.7	74.2±2.0	78.8±1.4	91.3±0.1	88.1±0.4	88.5±1.3	91.6±0.4	91.2±0.9	90.9±0.8
M^{3}BS (Ours)	**96.5±0.4**	**95.3±1.3**	**95.9±0.3**	**99.5±0.2**	**99.1±0.1**	**99.3±0.2**	**96.7±0.4**	**96.8±0.3**	**96.5±0.4**

same dataset. We use three well-known classification metrics for quantitative comparison: the overall accuracy (OA), the average accuracy (AA), and the Kappa coefficient (Kappa). The comprehensive results of all comparison algorithms on the three datasets are summarized in Table I, where each experiment was run 30 times independently to report the mean and standard deviation.

1) Indian Pines: For Indian Pines dataset, we randomly divided the samples into a 5% training set and a 95% testing set, selecting a subset of 20 bands from the original 200. As detailed in Indian Pines section of Table I, our M^{3}BS significantly outperforms all other methods. For instance, with an OA below 70%, BS-Nets and GCSR-BS show the weakest performance. Methods using auxiliary classifiers, such as mRMR and TGSR, perform slightly better but are hindered by the separate optimization of band selection and classification. In contrast, CNN-based algorithms demonstrate superior performance due to their strong spatial representation capabilities. Our M^{3}BS achieves the best accuracy across all three metrics, validating its effectiveness.

2) Pavia University: The dataset was split into 3% training samples and 97% testing samples, with 20 bands selected from the original 103. The results, shown in the Pavia section of Table I, follow a similar trend. End-to-end deep learning algorithms, including our M^{3}BS, achieve a 5%–15% accuracy improvement over traditional methods. M^{3}BS again obtains the best results, improving upon the strong SICNN baseline by 4.2% in OA, 6.0% in AA, and 5.6% in Kappa.

3) University of Houston: Following the same 5:95 training/testing split ratio as Indian Pines, we selected 20 bands from the 176 available bands. As presented in the Houston section of Table I, M^{3}BS performs best on most of the 15 categories, maintaining a classification accuracy above 94% on almost all of them. Compared with the competitive ABCNN method, M^{3}BS exhibits improvements of at least 5.1%, 5.6%, and 5.6% in OA, AA, and Kappa, respectively.

D. Zero-Shot Band Selection With Supervised Classification in a Cross-Scene Setting

We designed a cross-scene experiment to evaluate the zero-shot band selection capability of our proposed M^{3}BS framework. In this setting, the model must select informative bands for a new, entirely unseen dataset without using any of its samples during training. Specifically, we designated three additional hyperspectral datasets, namely, Salinas, Washington DC Mall, and KSC, as the meta-training tasks. The previously introduced Indian Pines, Pavia University, and University of Houston datasets served as the unseen meta-testing tasks. The M^{3}BS model was trained exclusively on the meta-training datasets to learn a generalizable band selection strategy. Subsequently, the trained model was directly applied to select a band subset from each unseen meta-testing dataset. To quantify the effectiveness of the selected bands, a supervised classifier was trained and evaluated using only the labeled samples from the corresponding meta-testing dataset. It is crucial to distinguish this scenario from unsupervised learning: in our zero-shot setting, the model has no access to any samples from the meta-testing datasets during training, whereas in unsupervised learning, samples are available but labels are not.

Some of the comparison algorithms are optimized for ZSL. To make training feasible, we apply some intuitive modifications to these algorithms. For multiple datasets to be trained on, we first find the dataset containing the most spectral bands, and then fill in the missing bands with 0 for other datasets with fewer bands. As a result, all training datasets are aligned in terms of the number of spectral bands. For mRMR, after calculating the score of the selected band combination for each training dataset, the average of these scores is taken to be the final score for further updates. For BS-Nets, we set up as many reconstruction networks as there are datasets, and take the average of the output of these networks as the final result. For CNN-based algorithms SICNN and ABCNN, their CNN-based classifiers are retrained for different band subsets during the training phase. For typical DA methods, CDWOASA and CSSRnet, these meta-training datasets are set up as SDs, and meta-testing scenes are utilized as TDs, respectively. To align with the DA framework, we configured the TD's split ratio following the methodology in Section IV-C, with its training samples subsequently employed for band selection.

It should be pointed out that, since our M^{3}BS is originally designed for ZSL, it is naturally compatible with multiple training datasets and does not require an additional dataset alignment operation. Taking the first graph convolutional layer as an example, the size $hw \times 256$ of the weight matrix $\mathbf{W}^{\mathrm{GCN}(1)}$ is independent of the number of bands n^{band} for any HSI dataset. As long as the patch size $h \times w$ is fixed,

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TABLE II
PERFORMANCE OF SUPERVISED CLASSIFICATION ON UNSEEN DATASETS USING BANDS SELECTED IN A ZERO-SHOT SETTING

Category	Method	Indian Pines			Pavia			Houston		
		OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa
Single Scene Methods	mRMR [67]	70.8±0.8	60.9±1.3	66.6±0.9	73.4±0.4	63.2±1.0	63.3±0.5	80.6±1.0	79.5±1.1	79.0±1.1
	BS-Nets [39]	74.0±0.3	64.3±1.6	70.0±0.3	79.5±0.4	66.7±1.1	71.2±0.6	79.7±0.5	79.2±0.4	78.1±0.5
	GCSR-BS [35]	70.5±1.4	65.5±2.6	66.2±1.6	82.6±0.3	76.6±0.7	76.2±0.4	88.5±1.2	87.4±1.2	87.6±1.2
	TGSR [68]	72.9±1.2	72.5±1.5	68.9±1.4	86.7±0.7	86.1±0.9	82.0±1.0	89.4±0.8	88.9±0.9	88.6±0.8
	DGLAnet [69]	78.6±0.8	76.9±1.7	76.4±1.3	87.3±0.5	86.1±0.7	86.2±0.7	88.6±0.2	87.0±0.5	87.9±0.6
	TSC [70]	69.3±2.6	63.5±2.4	63.6±1.4	79.7±1.2	77.6±1.1	77.3±0.9	81.4±0.6	81.1±0.2	80.7±0.3
	SICNN [31]	85.0±3.2	78.6±3.4	82.8±3.7	94.5±0.6	90.4±1.6	92.7±0.9	88.9±1.7	87.3±1.9	88.0±1.8
	ABCNN [33]	77.6±2.1	70.6±3.4	75.4±2.2	81.5±1.1	80.9±2.3	81.1±1.7	83.7±1.6	83.0±1.3	82.4±1.8
Homogeneous Cross-Scene Methods	CDWOASA [45]	72.6±2.0	72.2±2.0	68.6±2.3	84.8±1.2	84.5±1.2	79.5±1.7	88.3±1.1	87.8±1.1	87.3±1.2
	CSSRnet [46]	71.0±0.3	70.2±0.3	66.6±0.4	86.3±0.2	85.8±0.3	81.5±0.3	85.7±0.4	85.5±0.4	84.5±0.4
Heterogeneous Cross-Scene Methods	MR-Selection [22]	91.1±0.7	87.6±2.1	89.9±0.7	93.2±2.8	87.4±3.2	90.9±2.8	91.8±1.7	91.8±1.4	91.1±1.8
	M^3BS	**96.2±0.5**	**94.3±0.9**	**95.7±0.5**	**98.5±0.2**	**97.5±0.3**	**98.1±0.2**	**96.5±0.5**	**95.9±0.3**	**96.3±0.4**

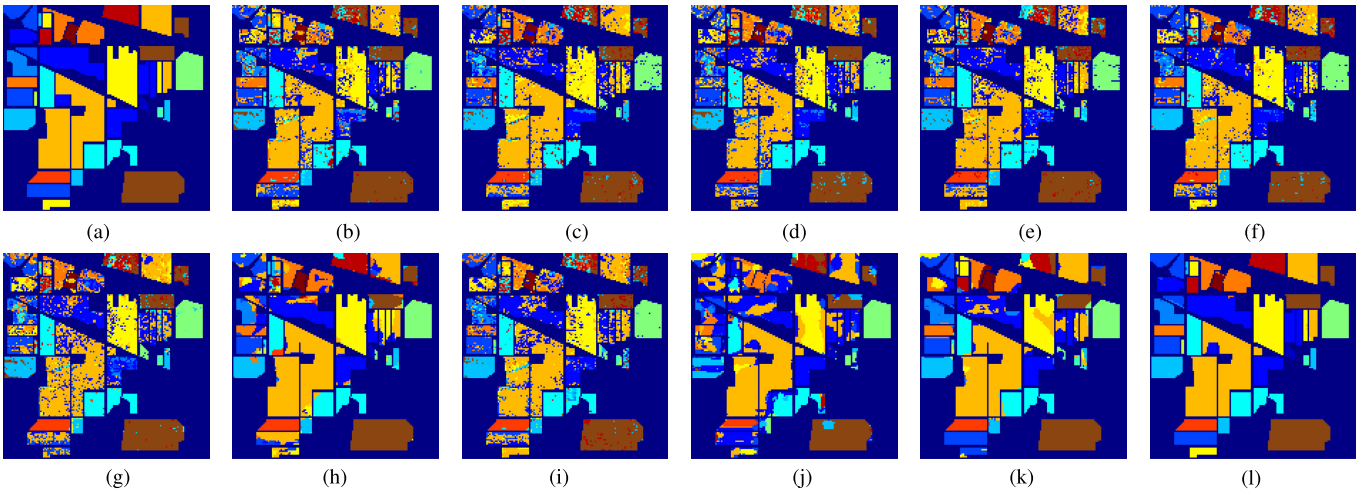


Fig. 6. Classification maps of M^3BS and competing methods on Indian Pines dataset under the zero-shot setting. (a) G.T. (b) mRMR. (c) BSNets. (d) GCSR-BS. (e) DRL-BS. (f) DGLAnet. (g) TSC. (h) SICNN. (i) CDWOASA. (j) CSSRnet. (k) MR-Selection. (l) M^3BS (ours).

the same learnable parameters are applicable to datasets with any number of bands. The relationship between the network structure and the number of bands is decoupled in M^3BS.

Table II shows the average of the classification metrics OA, AA, and Kappa, and their corresponding standard deviations on three datasets by 30 independent runs of each algorithm. Among these accuracies, the best ones are bolded and highlighted. It can be seen at a glance from these tables that our M^3BS significantly outperforms all the other nine algorithms, and maintains a consistently high level over all three datasets that are absent throughout the whole training phase. Quantitatively, M^3BS demonstrates a significant performance leap, with improvements on Indian Pines dataset reaching up to 5.1% in OA, 6.7% in AA, and 5.8% in Kappa over the strongest baseline. On other datasets, it also consistently maintains a clear advantage. In addition, M^3BS performs better than other zero-shot methods. This is predominantly because both the network structure and the optimization procedure in M^3BS are designed for adapting to zero-shot band selection. Specifically, the generalizable GCN with the additional supervision from multiple teachers can achieve meta-knowledge extraction that is compatible with various datasets. The multiobjective meta-learning optimization procedure makes M^3BS applicable to different objectives and different datasets. As a result, the extracted meta-knowledge can be directly transferred to unseen datasets without any retraining or fine-tuning. In contrast, the knowledge learned by other comparison algorithms is not transferable, or the transferability is poor, which means that these algorithms perform well on seen datasets but poorly on unseen datasets.

For a more intuitive comparison, the ground truth (G.T.) and the visual classification results on three unseen hyperspectral datasets of these algorithms are shown in Figs. 6–8. It should be noted that the full-image classification predictions of the University of Houston are given due to the scattered distribution of labeled pixel samples. As shown in Figs. 6(b)–(k)–8(b)–(k), a large number of pixel samples are misclassified into other categories. For mRMR, BS-Nets, GCSR-BS, TGSR, TSC, and CDWOASA, the network input is constructed at the pixel level, which does not consider

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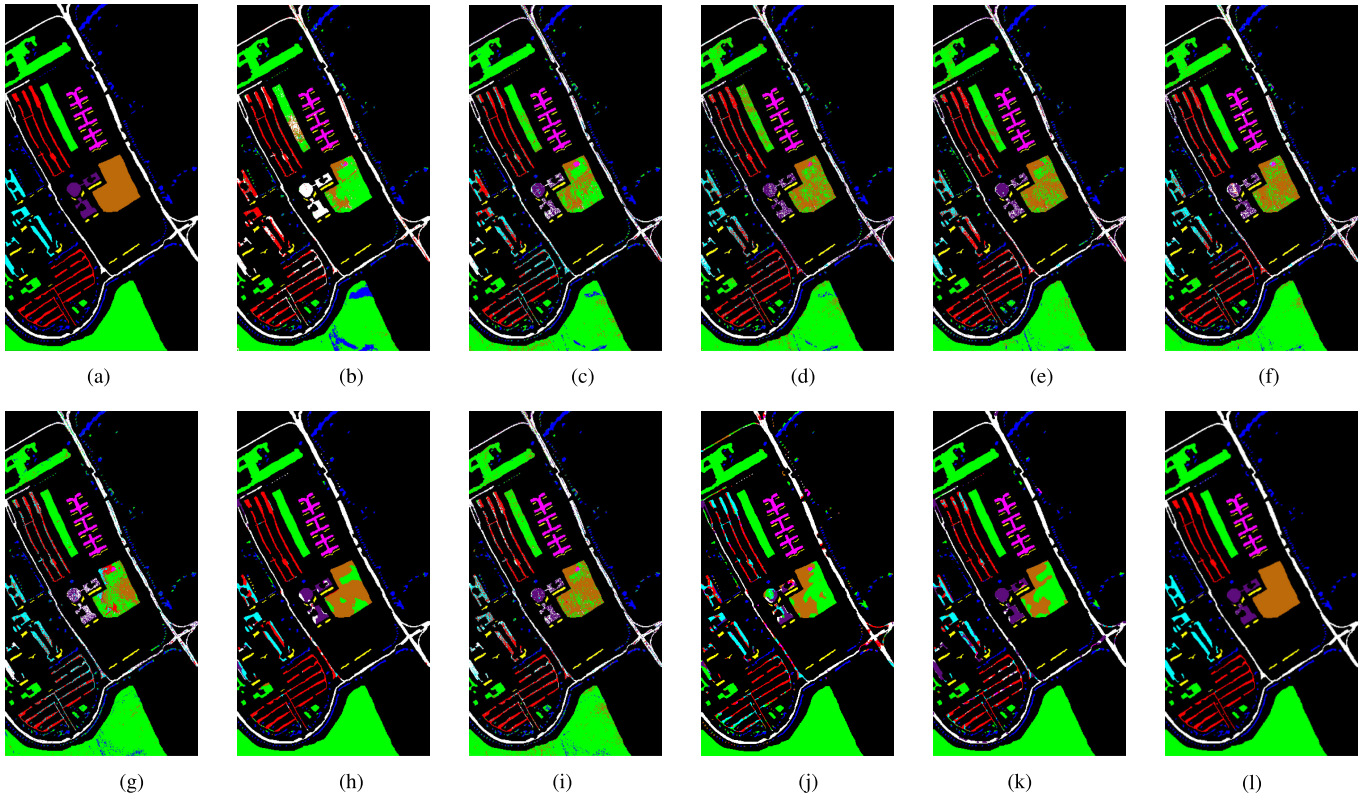


Fig. 7. Classification maps of M^{3}BS and competing methods on the Pavia University dataset under the zero-shot setting. (a) G.T. (b) mRMR. (c) BSNets. (d) GCSR-BS. (e) DRL-BS. (f) DGLAnet. (g) TSC. (h) SICNN. (i) CDWOASA. (j) CSSRnet. (k) MR-Selection. (l) M^{3}BS (ours).

the relationship among adjacent pixels and lacks local spatial information. As a result, the output of these methods contains lots of noise points and discontinuous areas, especially for categories like Grass-trees, Soybean-notill, soybean-mintill in Indian Pines and bare soil in Pavia University. For cross-scene algorithms CSSRnet and MR-Selection, the spatial network takes the spatial window around the pixel as input, which significantly improves local spatiality and category consistency in the same area. However, misclassified pixels still appear at some boundaries among different categories. In M^{3}BS, pixel-level samples are used in the process of constructing spatial–spectral graphs, and spatial windows are regarded as input for the classifier. Compared with the above methods, M^{3}BS not only utilizes a more efficient graph construction strategy, but also takes local spatial information into consideration. As a result, M^{3}BS obtains the best classification visualization among all comparison algorithms, which preserves detailed boundaries, has the fewest noise points, and is closest to the G.T.

E. Zero-Shot Band Selection With Zero-Shot Classification in a Cross-Scene Setting

While the previous section validated our zero-shot band selection using a supervised classifier, this experiment addresses a more challenging, fully end-to-end zero-shot scenario. Here, we evaluate the performance of M^{3}BS when coupled with a zero-shot classifier, requiring no labeled data from the target scene for either band selection or classification. To achieve this, we integrated M^{3}BS with the recently proposed SPECIAL [53] classifier, enabling joint, end-to-end optimization of both modules. The dataset configuration remains identical to the cross-scene setting described in Section IV-D. We also adapted the baseline methods for fair comparison. For pipeline-based methods (mRMR, TGSR, etc.), SPECIAL was used as a standalone classifier to evaluate their output. For other end-to-end methods (GCSR-BS, MR-Selection), we followed a similar joint-optimization approach. This fully zero-shot framework is made possible by leveraging CLIP's inherent generalization capabilities within the SPECIAL classifier. The model learns to classify unseen categories in the TD after being fine-tuned solely on SD data, completely avoiding the need for TD pretraining.

Table III shows the average of the classification metrics OA, AA, and Kappa, and their corresponding standard deviations on three datasets by 30 independent runs of each algorithm. Among these results, the best ones are bolded and highlighted. As these tables demonstrate, our M^{3}BS significantly outperforms all the other algorithms, and maintains a consistently high level over both datasets that are absent during the band selection and classification process. Quantitatively, M^{3}BS can improve at least 3.5% OA, 1.8% AA, and 3.9% Kappa compared to the best performing one among other algorithms. This is mainly because the multiobjective strategy is designed to adapt to ZSL and avoid overfitting to a single task. For mRMR, GCSR-BS, TGSR, DGLAnet, TSC, CDWOASA, and CSSRnet, the zero-shot classifier operates independently of band selection methods, which hinders mutual promotion between different tasks. In contrast to these methods, M^{3}BS introduces meta-optimized loss adaptation that

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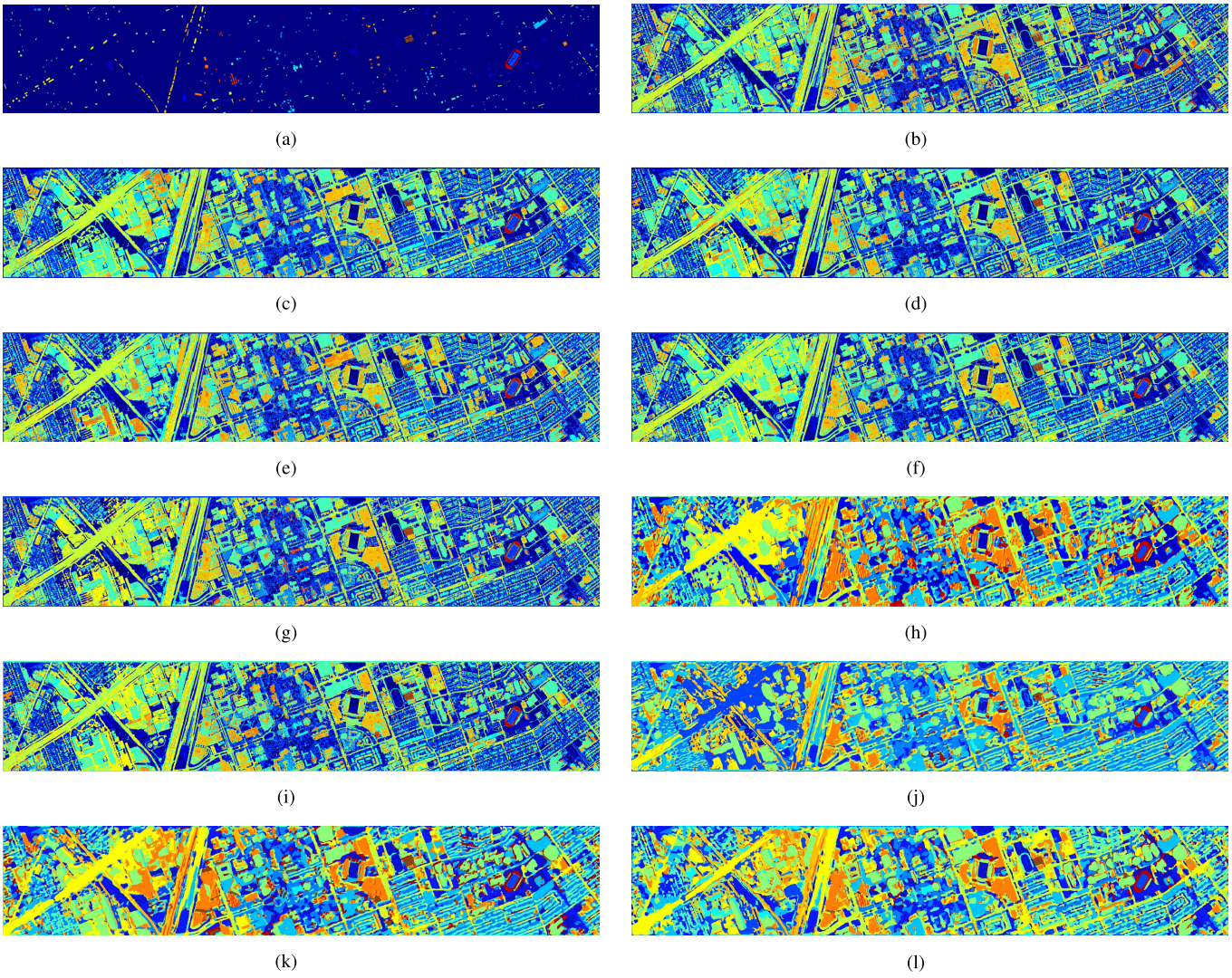


Fig. 8. Classification maps of M^{3}BS and competing methods on the University of Houston dataset under the zero-shot setting. (a) G.T. (b) mRMR. (c) BSNets. (d) GCSR-BS. (e) DRL-BS. (f) DGLAnet. (g) TSC. (h) SICNN. (i) CDWOASA. (j) CSSRnet. (k) MR-Selection. (l) M^{3}BS (ours).

TABLE III
PERFORMANCE FOR JOINT ZERO-SHOT BAND SELECTION AND CLASSIFICATION ON THE UNSEEN PAVIA UNIVERSITY DATASET

Category	Method	OA (%)	AA (%)	Kappa×100
Single-Scene Methods	mRMR [67]	51.6±0.8	42.2±1.3	42.7±0.9
	BS-Nets [39]	54.0±0.3	44.3±1.6	44.1±0.3
	GCSR-BS [35]	58.9±1.4	45.5±1.6	44.9±1.4
	TGSR [68]	56.2±0.1	48.3±0.1	45.9±0.1
	DGLAnet [69]	59.8±0.8	49.7±0.7	46.0±1.3
	TSC [70]	55.2±0.6	43.5±1.4	44.4±0.4
	SICNN [31]	60.8±1.7	49.4±0.7	50.7±1.8
	ABCNN [33]	59.3±1.7	49.7±1.0	49.9±0.7
Homogeneous Cross-Scene	CDWOASA [45]	58.6±1.0	50.1±0.2	49.4±1.2
	CSSRnet [46]	59.5±2.1	47.3±0.8	43.3±1.2
Heterogeneous Cross-Scene	MR-Selection [22]	60.3±0.4	49.9±0.1	50.2±0.9
	M^{3}BS (Ours)	**64.3±0.2**	**51.9±0.7**	**54.6±0.9**

dynamically balances band selection and zero-shot discriminative objectives, thereby significantly outperforming other end-to-end methods. The integration of M^{3}BS with a hyperspectral zero-shot classifier demonstrates that the selected band subset enables the classifier to acquire the transferable meta-knowledge, leveraging M^{3}BS's effectiveness in zero-shot band selection.

F. Ablation Study

In this section, we conduct several ablation studies to validate the effectiveness of the key components in M^{3}BS. All experiments are performed under the "zero-shot band selection with supervised classification in a cross-scene setting," as detailed in Section IV-D. Performance is evaluated on the three unseen meta-testing datasets (Indian Pines, Pavia University, and University of Houston) using the standard OA, AA, and Kappa metrics.

1) Multiple Teachers: This ablation aims to verify whether multiple teachers are beneficial for band selection. To quantitatively analyze the contribution of each teacher, Table IV records the classification results of M^{3}BS with different numbers of teachers.

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TABLE IV
CLASSIFICATION RESULTS OF M^{3}BS WITH DIFFERENT NUMBERS OF TEACHERS ON DIFFERENT DATASETS

Dataset	Condition	OA (%)	AA (%)	Kappa×100
Indian Pines	M^{3}BS w/o teachers	93.9	90.3	93.1
	M^{3}BS w/ 1 teacher (BS-Nets)	95.6	91.4	95.0
	M^{3}BS w/ 2 teachers (BS-Nets and SICNN)	95.8	92.3	95.2
	M^{3}BS w/ 3 teachers (BS-Nets, SICNN and TWCNN)	**96.2**	**94.3**	**95.7**
Pavia University	M^{3}BS w/o teachers	97.2	95.7	96.4
	M^{3}BS w/ 1 teacher (BS-Nets)	98.2	96.7	97.7
	M^{3}BS w/ 2 teachers (BS-Nets and SICNN)	98.4	97.1	97.9
	M^{3}BS w/ 3 teachers (BS-Nets, SICNN and TWCNN)	**98.5**	**97.5**	**98.1**
University of Houston	M^{3}BS w/o teachers	93.4	92.9	93.0
	M^{3}BS w/ 1 teacher (BS-Nets)	95.0	94.5	94.7
	M^{3}BS w/ 2 teachers (BS-Nets and SICNN)	95.9	95.3	95.6
	M^{3}BS w/ 3 teachers (BS-Nets, SICNN and TWCNN)	**96.5**	**95.9**	**96.3**

TABLE V
CLASSIFICATION RESULTS OF M^{3}BS WITH DIFFERENT MULTITEACHER FUSION STRATEGIES ON DIFFERENT DATASETS

Dataset	Condition	OA (%)	AA (%)	Kappa×100
Indian Pines	union	95.4	92.4	94.8
	normalized sum	96.1	91.7	95.5
	diversity ensemble	**96.2**	**94.3**	**95.7**
Pavia University	union	98.0	96.7	97.4
	normalized sum	98.2	96.8	97.7
	diversity ensemble	**98.5**	**97.5**	**98.1**
University of Houston	union	96.4	**96.1**	96.2
	normalized sum	92.7	92.6	92.2
	diversity ensemble	**96.5**	95.9	**96.3**

As shown in Table IV, as the number of band selection teachers increases, the classification performance of M^{3}BS keeps improving. The number of teachers being zero corresponds to the absence of both the band selection loss $\mathcal{L}^{\text{bs}}$ and the multiobjective loss $\mathcal{L}$. At this time, the classification accuracy is about 3% lower than the original M^{3}BS, which proves that the classification task alone cannot provide sufficient meta-knowledge for zero-shot band selection. As different teachers join, the classification ability becomes stronger and stronger since different types of teachers can provide guidance from different perspectives. In addition, the order in which teachers join shows that the supervision from deep learning algorithms like SICNN and TWCNN is of higher quality than that from unsupervised algorithms like BS-Nets.

2) Multiteacher Fusion: After confirming that multiple teachers are indeed beneficial for zero-shot band selection, this ablation tries to figure out the best strategy to fuse these teachers into one integrated teacher.

Our original M^{3}BS utilizes a strategy called diversity ensemble, which chooses the spectral bands according to their popularity. Bands with more received votes will be selected first, and bands with the same number of received votes will be randomly selected. For comparison, we compare M^{3}BS with two intuitive fusion strategies, named union and normalized sum. The former treats the band combinations selected by teachers as sets, and takes the union of these sets as the result. The latter multiplies each teacher's output score by one-third and adds these scores together as the ground truth.

Table V records the classification results of M^{3}BS with different multiteacher fusion strategies. As shown in Table V, our diversity ensemble strategy consistently achieves the best or near-best results. Compared to other fusion methods, it provides a modest but stable performance gain on datasets like Pavia University, while demonstrating a more significant advantage of up to 3.8% in OA on the University of Houston dataset. For the strategy union, we suspect that this is due to the so-called over-selection problem. In other words, when the bands selected by different teachers do not have much overlap, too many bands will be used as fusion results, and some bands with low discriminability may be added, which in turn downgrades the classification performance. For the strategy normalized sum, we think that it may lack the introduction of random noise, thus leaving the supervision for each epoch unchanged and reducing the generalization ability. On the contrary, the diversity ensemble strategy we proposed not only limits the number of selected bands but also introduces a certain degree of randomness, thereby ensuring the effective fusion of multiple teachers.

3) Analysis of Multiobjective Loss Weighting: This ablation is used to prove the necessity of balancing multiple losses, and to make a comparison between hand-crafted and automatically learned loss weights. When encountering multiple loss functions in a deep learning algorithm, the intuitive idea is to manually set up different weights for different losses and keep these weights unchanged throughout the training process. Here, we design six hand-crafted weight combinations in sequence. Also, as usual, the sum of all the loss weights in a single combination is fixed to 1. Instead of manual intervention on these weights, our M^{3}BS utilizes a homoscedastic uncertainty-based multiobjective loss to automatically learn them. For a more refined and convincing comparison, Table VI elaborately lists the classification results of M^{3}BS with different multiloss weighting schemes.

As shown in Table VI, the uncertainty-based weighting scheme introduced in M^{3}BS dominates all other weighting schemes. First, we do not show the classification performance when only optimizing the band selection loss $\mathcal{L}^{\text{bs}}$. When the classification loss $\mathcal{L}^{\text{cls}}$ is absent, the CNN-based classifier of each dataset is not trained anymore, leading to random network parameters for inferring, which makes no sense. Next, the

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TABLE VI

CLASSIFICATION RESULTS OF M^{3}BS WITH DIFFERENT MULTILOSS WEIGHTING SCHEMES ON DIFFERENT DATASETS. (IP: INDIAN PINES, PU: PAVIA UNIVERSITY, AND UH: UNIVERSITY OF HOUSTON)

Scenario	Dataset	Weights λ^{bs}	Weights λ^{cls}	OA (%)	AA (%)	Kappa×100
only $\mathcal{L}^{cls}$	IP	0	1	93.9	90.3	93.1
	PU	0	1	97.2	95.7	96.4
	UH	0	1	93.4	92.9	93.0
hand-crafted weighting	IP	0.01	0.99	95.5	92.2	94.9
	PU	0.01	0.99	98.1	96.8	97.6
	UH	0.01	0.99	95.0	94.3	94.7
	IP	0.05	0.95	95.7	92.1	95.2
	PU	0.05	0.95	98.0	96.7	97.5
	UH	0.05	0.95	94.9	94.6	94.6
	IP	0.1	0.9	95.7	93.0	95.2
	PU	0.1	0.9	98.0	96.6	97.5
	UH	0.1	0.9	95.0	94.3	94.6
	IP	0.3	0.7	96.0	93.1	95.5
	PU	0.3	0.7	98.2	97.1	97.8
	UH	0.3	0.7	95.1	94.5	94.8
	IP	0.7	0.3	96.2	93.3	95.7
	PU	0.7	0.3	98.2	96.9	97.7
	UH	0.7	0.3	95.0	94.5	94.7
	IP	0.9	0.1	95.9	92.5	95.3
	PU	0.9	0.1	98.2	96.8	97.7
	UH	0.9	0.1	95.0	94.5	94.7
averaged weighting	IP	0.5	0.5	95.6	92.3	95.0
	PU	0.5	0.5	98.3	97.0	97.9
	UH	0.5	0.5	95.0	94.4	94.7
uncertainty weighting	IP	auto		**96.2**	**94.3**	**95.7**
	PU			**98.5**	**97.5**	**98.1**
	UH			**96.5**	**95.9**	**96.3**

case of only optimizing the classification loss $\mathcal{L}^{cls}$ is actually the same as the case without multiple teachers in Table IV, and the accuracy is about 3% lower than M^{3}BS. Here, the same assertion can be applied: a single classification task is not enough for sufficient meta-knowledge for zero-shot band selection. Among these hand-crafted weight combinations, we can observe that for the datasets Indian Pines, Pavia University, and University of Houston, the optimal weight ratios are approximately 0.7:0.3, 0.5:0.5, and 0.3:0.7, respectively. Finally, the weight combination learned by our uncertainty weighting scheme is better than any hand-crafted weight. It is reasonable to speculate that, since these weights are updated together with the network parameters, they can be adapted to different training stages. For example, in the early stage, band selection is entrusted to dominate the network training, leading to a preliminary understanding of spectral bands; in the later stage, the classification takes the lead to further strengthen the feature representation ability. This is only available for multiobjective learning, not for static weights.

G. Analysis of the Selected Spectral Bands

To show the selected bands more intuitively, a fixed number of 20 spectral bands is selected by different algorithms on three HSI datasets. In Fig. 9, the upper part of each subfigure displays the distribution of selected bands, while the lower part presents the information entropy curve of the entire spectrum.

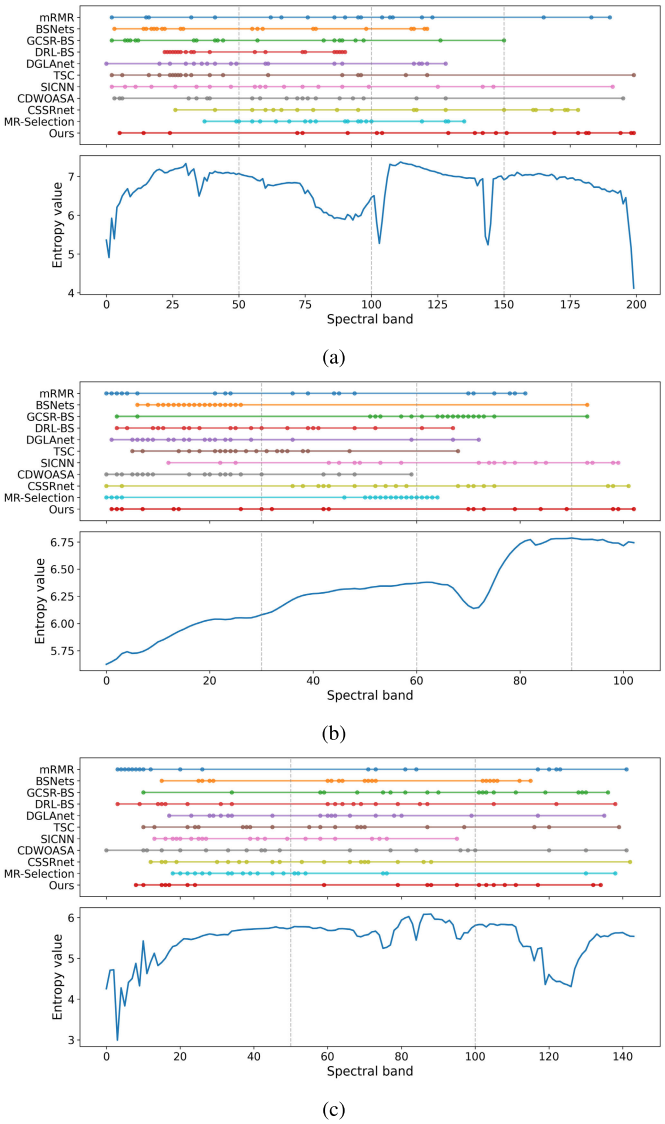


Fig. 9. Visualization of the selected bands (upper part) and information entropy (lower part) on three datasets under the zero-shot band selection with zero-shot classification in a cross-scene setting. M^{3}BS selects discrete bands in high-entropy regions, avoiding redundancy and noise. (a) Indian Pines. (b) Pavia University. (c) University of Houston.

As shown in Fig. 9, the bands selected by CDWOASA and CSSRnet exhibit a denser distribution. Since adjacent bands in HSIs typically share high correlation, such continuous selection results in significant information redundancy. Regarding information content, spectral bands with lower entropy values generally contain less information and are more likely to be noisy. It can be observed that methods like TSC and DGLAnet select several bands located in low-entropy valleys. In contrast, our proposed M^{3}BS successfully avoids these low-entropy noisy regions and selects bands with a more discrete distribution concentrated in high-entropy areas, thereby balancing redundancy reduction and information preservation.

V. CONCLUSION

In this article, a novel multiteacher multiobjective meta-learning framework, M^{3}BS, is proposed for zero-shot hyperspectral band selection. For compatible meta-knowledge

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extraction for various datasets, M^{3}BS constructs a generalizable GCN with spatial–spectral graphs. The decomposition of dataset-agnostic bases and dataset-specific coefficients makes it possible to explicitly characterize unseen samples with the unified meta-knowledge. For a better generalization ability and a faster convergence speed, M^{3}BS ensembles multiple teachers to provide high-quality experiences. For zero-shot optimization over multiple datasets and multiple objectives simultaneously, all these modules are integrated into a multiobjective meta-learning procedure. Experimental results over three commonly used hyperspectral datasets prove that M^{3}BS can produce better band combinations than many other state-of-the-art baselines for ZSL. Moreover, it provides a new design paradigm for subsequent researchers, and reminds them to pay more attention to the performance when transferring to unseen datasets.

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